

**Started on** Sunday, 26 April 2026, 10:04 AM

**State** Finished

**Completed on** Sunday, 26 April 2026, 12:03 PM

**Time taken** 1 hour 58 mins

**Grade** 7.00 out of 10.00 (70%)

**Question 1** | Correct Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

**Input Format:**

The first line contains S1.

The second line contains S2.

The third line contains N.

**Output Format:**

The first line contains the N characters present in S1 which are also present in S2.

**Boundary Conditions:**

$2 \leq N \leq 10$

$2 \leq \text{Length of } S1, S2 \leq 1000$

**Example Input/Output 1:**

Input:

```
abcbde
cdefghbb
3
```

Output:

```
bcd
```

**Note:**

b occurs twice in common but must be printed only once.

**Answer:** (penalty regime: 0 %)

```
1 s1 = input()
2 s2 = input()
3 n = int(input())
4
5 chars = set(s2)
6 res = []
7 seen = set()
8
9 for c in s1:
10     if c in chars and c not in seen:
11         res.append(c)
12         seen.add(c)
13     if len(res) == n:
```

```
13 |         if len(res) == 1:
14 |             break
15 |
16 |     print(' '.join(res))
17 |
```

|   | Input                   | Expected | Got |   |
|---|-------------------------|----------|-----|---|
| ✓ | abcbde<br>cdefghbb<br>3 | bcd      | bcd | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 2** | Incorrect Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

**Sample Input 1**

Malayalam is my mother tongue

**Sample Output 1**

is my mother tongue

**Answer:** (penalty regime: 0 %)

```

1 import string
2 s = input()
3 res = []
4 for w in s.split():
5     t = w.lower()
6
7     while t and not t[0].isalpha(): t = t[1:]
8     while t and not t[: -1].isalpha(): t = t[: -1]
9
10    if t != t[::-1] or not t:
11        if t: res.append(w)
12 print(" ".join(res))

```

|   | Input                         | Expected            | Got                 |   |
|---|-------------------------------|---------------------|---------------------|---|
| ✓ | Malayalam is my mother tongue | is my mother tongue | is my mother tongue | ✓ |

Your code failed one or more hidden tests.

Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLRLRL

Output:

4

Explanation: s can be split into "RL", "RRL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

| Test                                 | Result |
|--------------------------------------|--------|
| print(BalancedStrings('RLRRLRLRL'))  | 4      |
| print(BalancedStrings('RLLLLRRRLR')) | 3      |

Answer: (penalty regime: 0 %)

Reset answer

```
1 def BalancedStrings(s):
2     balance = 0
3     count = 0
4     for ch in s:
5         if ch == 'L':
6             balance += 1
7         else:
8             balance -= 1
```

```
9 |         if balance == 0:
10 |             count += 1
11 |     return count
```

|   | Test                                 | Expected | Got |   |
|---|--------------------------------------|----------|-----|---|
| ✓ | print(BalancedStrings('RLRLLRLRL'))  | 4        | 4   | ✓ |
| ✓ | print(BalancedStrings('RLLLLRRRLR')) | 3        | 3   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 4** | Correct Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

**Input Format**

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

**Output Format**

Print Yes or No.

**Sample Input**

```
7
0 1 2 4 6 5 3
1
```

**Sample Output**

Yes

**For example:**

| Input                       | Result |
|-----------------------------|--------|
| 5<br>8 9 12 15 3<br>11      | Yes    |
| 6<br>2 9 21 32 43 43 1<br>4 | No     |

**Answer:** (penalty regime: 0 %)

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 k = int(input())
4
5 d = {}
6 for num in arr:
7     if k - num in d:
8         print("Yes")
9         break
10    d[num] = d.get(num,0) + 1
11 else:
12    print("No")
```

|   | Input                       | Expected | Got |   |
|---|-----------------------------|----------|-----|---|
| ✓ | 5<br>8 9 12 15 3<br>11      | Yes      | Yes | ✓ |
| ✓ | 6<br>2 9 21 32 43 43 1<br>4 | No       | No  | ✓ |
| ✓ | 6<br>13 42 31 4 8 9<br>17   | Yes      | Yes | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



**Question 5** | Correct Mark 1.00 out of 1.00

Write a Python program for binary search.

**For example:**

| Input             | Result |
|-------------------|--------|
| 1,2,3,5,8<br>6    | False  |
| 3,5,9,45,42<br>42 | True   |

**Answer:** (penalty regime: 0 %)

```

1 arr = sorted(list(map(int, input().split(','))))
2 x = int(input())
3 l,r =0,len(arr) -1
4 found = False
5 while l <=r:
6     m = (l +r) //2
7     if arr[m] ==x:
8         found = True
9         break
10    if arr[m] < x:
11        l = m + 1
12    else:
13        r = m - 1
14 print(found)

```

|   | Input                | Expected | Got   |   |
|---|----------------------|----------|-------|---|
| ✓ | 1,2,3,5,8<br>6       | False    | False | ✓ |
| ✓ | 3,5,9,45,42<br>42    | True     | True  | ✓ |
| ✓ | 52,45,89,43,11<br>11 | True     | True  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 6** | Correct Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

**Constraints**

1<= string length <= 200

**Sample Input 1**

experience  
enc

**Sample Output 1**

xpri

**Answer:** (penalty regime: 0 %)

```
1 s1 = input()
2 s2 = input()
3 remove = set(s2)
4 res = ''.join([c for c in s1 if c not in remove])
5 print(res)
```

|   | Input             | Expected | Got  |   |
|---|-------------------|----------|------|---|
| ✓ | experience<br>enc | xpri     | xpri | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 7** | Correct Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array*.

**Example 1:**

**Input:** `nums = [3,0,1]`

**Output:** 2

**Explanation:** `n = 3` since there are 3 numbers, so all numbers are in the range `[0,3]`. 2 is the missing number in the range since it does not appear in `nums`.

**Example 2:**

**Input:** `nums = [0,1]`

**Output:** 2

**Explanation:** `n = 2` since there are 2 numbers, so all numbers are in the range `[0,2]`. 2 is the missing number in the range since it does not appear in `nums`.

**Example 3:**

**Input:** `nums = [9,6,4,2,3,5,7,0,1]`

**Output:** 8

**Explanation:** `n = 9` since there are 9 numbers, so all numbers are in the range `[0,9]`. 8 is the missing number in the range since it does not appear in `nums`.

**For example:**

| Test                                       | Result |
|--------------------------------------------|--------|
| <code>print(missingNumber([3,0,1]))</code> | 2      |
| <code>print(missingNumber([0,1]))</code>   | 2      |

**Answer:** (penalty regime: 0 %)

[Reset answer](#)

```
1 def missingNumber(nums):
2     n = len(nums)
3     return n * (n + 1) // 2 - sum(nums)
4
```

|   | Test                                      | Expected | Got |   |
|---|-------------------------------------------|----------|-----|---|
| ✓ | print(missingNumber([3,0,1]))             | 2        | 2   | ✓ |
| ✓ | print(missingNumber([0,1]))               | 2        | 2   | ✓ |
| ✓ | print(missingNumber([9,6,4,2,3,5,7,0,1])) | 8        | 8   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 8** | Correct Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element  $a[i]$  is a peak element if

$A[i-1] \leq A[i] \geq a[i+1]$  for middle elements.  $[0 < i < n-1]$

$A[i-1] \leq A[i]$  for last element  $[i=n-1]$

$A[i] \geq A[i+1]$  for first element  $[i=0]$

**Input Format**

The first line contains a single integer  $n$ , the length of  $A$ .

The second line contains  $n$  space-separated integers,  $A[i]$ .

**Output Format**

**Print** peak numbers separated by space.

**Sample Input**

5

8 9 10 2 6

**Sample Output**

10 6

**For example:**

| Input    | Result |
|----------|--------|
| 4        | 12 8   |
| 12 3 6 8 |        |

**Answer:** (penalty regime: 0 %)

```

1 n= int(input())
2 a = list(map(int, input().split()))
3 res = []
4
5 for i in range (n):
6     if n ==1:
7         res.append(a[i])
8     elif i ==0:
9         if a[i] >= a[i+1]:
10            res.append(a[i])
11    elif i == n-1:
12        if a[i] >= a[i -1]:
13            res.append(a[i])
14    else:
15        if a[i - 1] <= a[i] >=a[i + 1]:
16            res.append(a[i])
17
18 print(*res)
19

```

|   | Input                | Expected  | Got       |   |
|---|----------------------|-----------|-----------|---|
| ✓ | 7<br>15 7 10 8 9 4 6 | 15 10 9 6 | 15 10 9 6 | ✓ |
| ✓ | 4<br>12 3 6 8        | 12 8      | 12 8      | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 9** | Incorrect Mark 0.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`.

If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with  $O(\log n)$  runtime complexity.

**Example 1:**

**Input:** `nums = [-1,0,3,5,9,12]`, `target = 9`

**Output:** `4`

**Explanation:** 9 exists in `nums` and its index is 4

**Example 2:**

**Input:** `nums = [-1,0,3,5,9,12]`, `target = 2`

**Output:** `-1`

**Explanation:** 2 does not exist in `nums` so return -1

**Constraints:**

- $1 \leq \text{nums.length} \leq 10^4$
- $-10^4 < \text{nums}[i], \text{target} < 10^4$
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

**For example:**

| Test                                          | Result |
|-----------------------------------------------|--------|
| <code>print(search([-1,0,3,5,9,12],9))</code> | 4      |

**Answer:** (penalty regime: 0 %)

Reset answer

```

1 def search(nums, target):
2     l,r = 0,len(nums) -1
3     while l <+ r:
4         m = (1 + r) //2
5         if nums[m] == target:
6             return m
7         if nums[m] < target:
8             l = m + 1
9         else:
10            r = m - 1
11     return -1
12
13

```

**Syntax Error(s)**

File "\_\_tester\_\_.python3", line 8

```
1 = m + 1
```

```
^
```

SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?

**Incorrect**

Marks for this submission: 0.00/1.00.



Question 10

Incorrect

Mark 0.00 out of 1.00

You are given an  $m \times n$  integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` if `target` is in `matrix` or `False` otherwise.

You must write a solution in  $O(\log(m * n))$  time complexity.

Example 1:

|    |    |    |    |
|----|----|----|----|
| 1  | 3  | 5  | 7  |
| 10 | 11 | 16 | 20 |
| 23 | 30 | 34 | 60 |

Input: `matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, `target = 3`

Output: `True`

Example 2:

|    |    |    |    |
|----|----|----|----|
| 1  | 3  | 5  | 7  |
| 10 | 11 | 16 | 20 |
| 23 | 30 | 34 | 60 |

Input: `matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, `target = 13`

Output: `False`

For example:

| Test                                                                          | Result |
|-------------------------------------------------------------------------------|--------|
| <code>print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))</code> | False  |
| <code>print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))</code>  | True   |

Answer: (penalty regime: 0 %)

Reset answer

```
1 def searchMatrix(matrix, target):
2     if not matrix or not matrix[0]:
3         return False
4
5     m,n = len(matrix), len(matrix[0])
6     left, right = 0, m * n - 1
```

```
6 | l, r = 0, m - 1
7 |
8 | while l <= r:
9 |     mid = (l + r) // 2
10 |    val = matrix[mid // n][mid % n]
11 |
12 |    if val == target:
13 |        return True
14 |    elif val < target:
15 |        l = mid + 1
16 |    else:
17 |        r = mid - 1
18 |    return False
```

|   | Test                                                             | Expected | Got  |   |
|---|------------------------------------------------------------------|----------|------|---|
| ✖ | print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13)) | False    | None | ✖ |
| ✖ | print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))  | True     | None | ✖ |

Your code must pass all tests to earn any marks. Try again.

Show differences

Incorrect

Marks for this submission: 0.00/1.00.